

Customer Shopping Behavior Analysis

Revised edition — corrected segmentation, data-backed recommendations, added SQL techniques

Executive Summary

This analysis covers 3,900 retail transactions to identify what actually drives spend and engagement. The headline finding is a negative one worth stating plainly: subscription status, customer tenure, and age group all show almost no difference in average spend per order (roughly \$59–60 across every cut). The clearest real signal is seasonal timing — Fall outperforms Summer by about 7.7% in average order value — alongside product category, where Footwear and Clothing carry above-average order values and Outerwear lags behind. Recommendations in Section 6 are built directly from these findings.

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key features: customer demographics (Age, Gender, Location, Subscription Status); purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color); shopping behavior (Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing data: 37 values in the Review Rating column
- No transaction date/year field exists — only a categorical Season field, used as the closest available trend proxy

3. Data Preparation (Python)

The steps below are shown as they were actually run in the notebook — code cell followed by its real output — rather than summarized as bullet points.

Loading the data

Loaded the raw CSV with pandas and confirmed the structure with `df.head()` — 3,900 rows, 18 columns, one row per transaction.

```
df = pd.read_csv('customer_shopping_behavior.csv')
df.head()
```

customer_id	age	gender	item_purchased	category	purchase_amount	location
1	55	Male	Blouse	Clothing	53	Kentucky
2	19	Male	Sweater	Clothing	64	Maine
3	50	Male	Jeans	Clothing	73	Massachusetts
4	21	Male	Sandals	Footwear	90	Rhode Island
5	45	Male	Blouse	Clothing	49	Oregon

Missing data handling

`df.isnull().sum()` showed 37 missing values, all in Review Rating. These were imputed with the median rating within the same product category, rather than the column-wide median, so ratings stay realistic for each category.

```
# Check for missing values
df.isnull().sum()
```

Column	Missing Values
review_rating	37
all other columns	0

Feature engineering — age_group

Customers were split into four age groups using quartile bins (`pd.qcut`), so each group holds roughly the same number of customers rather than fixed age ranges.

```
# Age group – quartile bins so each group is a comparable share of customers
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
df[['age', 'age_group']].head(6)
```

age	age_group
55	Middle-aged
19	Young Adult
50	Middle-aged
21	Young Adult
45	Middle-aged
46	Middle-aged

Corrected customer segmentation

The original New/Returning/Loyal thresholds were fixed numbers that put about 80% of customers into a single “Loyal” bucket. Re-derived using the 25th and 75th percentile of previous_purchases (13 and 38) instead, so the three segments are actually comparable in size.

```
# Segment customers using 25th/75th percentile of previous_purchases (13, 38)
def segment_customer(p):
    if p <= 13: return 'New'
    elif p <= 38: return 'Returning'
    else: return 'Loyal'

df['customer_segment'] = df['previous_purchases'].apply(segment_customer)
df['customer_segment'].value_counts()
```

customer_segment	count
Returning	1958
New	1014
Loyal	928

Also verified that discount_applied and promo_code_used were fully redundant (identical on every row) before dropping the latter, and connected the cleaned DataFrame to PostgreSQL for the SQL analysis in Section 4.

4. Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions. Corrected and newly added queries are noted where relevant.

1. Revenue by Gender

Gender	Revenue (\$)
Female	75,191
Male	157,890

5. Subscribers vs. Non-Subscribers

Subscription	Customers	Avg Spend (\$)	Total Revenue (\$)
No	2,847	59.87	170,436
Yes	1,053	59.49	62,645

Finding: subscribers spend slightly less per order on average than non-subscribers. Subscription status is not a meaningful predictor of order value — the lower total revenue from subscribers is purely a function of there being fewer of them (1,053 vs. 2,847), not lower engagement per subscriber.

7. Customer Segmentation (Corrected)

Segment	Customers	Avg Spend (\$)	Total Revenue (\$)
Returning	1,958	59.47	116,444
New	1,014	60.14	60,980
Loyal	928	59.98	55,657

Finding: with balanced segment sizes (using the 25th/75th percentile of previous purchases, instead of fixed thresholds), average spend is nearly identical across New, Returning, and Loyal customers (\$59–60). Tenure alone does not predict order value.

11. Revenue by Season (new)

Season	Orders	Total Revenue (\$)	Avg Order Value (\$)
Fall	975	60,018	61.56
Spring	999	58,679	58.74
Winter	971	58,607	60.36
Summer	955	55,777	58.41

Finding: Fall shows the highest average order value, about 7.7% above Summer's — the clearest actionable pattern in the dataset, since it holds even though subscription and segment status don't predict spend.

12. Categories Above Average Order Value (new — HAVING clause)

Category	Avg Order Value (\$)	Orders
Footwear	60.26	599
Clothing	60.03	1,737
Accessories	59.84	1,240

Outerwear (324 orders) is the only category below the overall average — worth investigating for a pricing, sizing, or seasonality issue.

3. Top 5 Products by Average Rating

Item Purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

6. Top 5 Discount-Dependent Products

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

13. Locations with Elevated Low-Rating Rates (new — correlated subquery)

Location	Low-Rating Rate (%)
Oklahoma	28.00
West Virginia	25.93
Connecticut	25.64
Georgia	24.05
Arizona	23.08

Locations shown have a rate of ratings below 3.0 higher than the overall average.

8. Top 3 Products per Category

Rank	Category	Item Purchased	Orders
1	Accessories	Jewelry	171

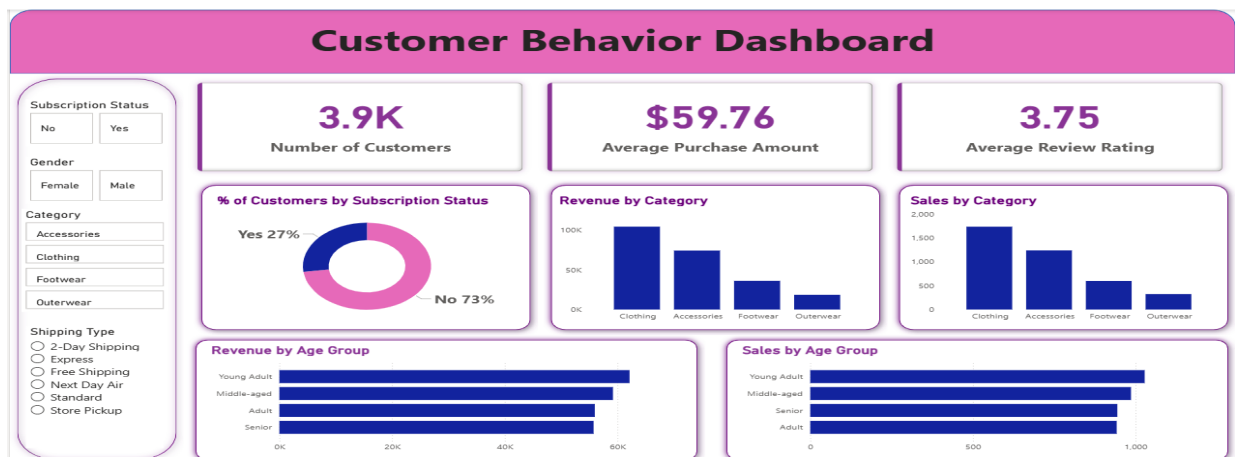
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

10. Revenue by Age Group

Age Group	Total Revenue (\$)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

5. Dashboard in Power BI

An interactive dashboard was built in Power BI to present insights visually, with filters for subscription status, gender, category, and shipping type.



6. Business Recommendations

Each recommendation below is tied directly to a query result above, rather than restating generic retail best practice. The two Python sanity checks below (run before the SQL analysis) back the two most counter-intuitive findings.

Sanity check — subscribers vs. non-subscribers

Grouped directly in pandas to confirm the SQL result before trusting it: subscribers spend essentially the same per order as non-subscribers.

```
# Subscribers vs non-subscribers: does subscribing mean higher spend?
df.groupby('subscription_status')['purchase_amount'].agg(['count', 'mean', 'sum'])
```

subscription_status	count	mean	sum
No	2847	59.87	170436
Yes	1053	59.49	62645

Sanity check — revenue by season

Since the dataset has no date column, season is the only available trend proxy. Fall comes out clearly ahead on both total revenue and average order value.

```
# Revenue by season (closest available trend proxy – no date column exists)
df.groupby('season')['purchase_amount'].sum().sort_values(ascending=False)
```

season	purchase_amount
Fall	60018
Spring	58679
Winter	58607
Summer	55777

- **Time seasonal promotions around Fall, not loyalty tier.**

Fall has the highest average order value (7.7% above Summer). Since neither subscription status nor customer segment predicts spend, seasonal timing is the strongest lever the data actually supports for driving order value up.

- **Investigate Outerwear's below-average performance.**

Outerwear is the only category with an average order value below the overall mean and has the lowest order volume (324). Worth checking pricing, sizing options, or whether it's under-promoted relative to Clothing and Footwear.

- **Reconsider subscription and loyalty-tier messaging.**

Subscribers do not out-spend non-subscribers, and Loyal customers do not out-spend New customers. A subscription/loyalty push should be framed around purchase frequency or retention, not order value, since the data doesn't support that framing.

- **Review discount policy on the five discount-dependent items.**

Hat, Sneakers, Coat, Sweater, and Pants are discounted in roughly half of all purchases. Confirm these discounts are driving incremental volume rather than subsidizing sales that would have happened anyway.

- **Address markets with elevated low-rating rates.**

Oklahoma, West Virginia, and Connecticut show the highest share of sub-3.0 review ratings. Worth checking for a regional fulfillment, sizing, or product-mix issue.

7. Deliverables

- customer_shopping_behavior_cleaned.csv — cleaned dataset with engineered features
- Customer_Shopping_Behavior_Analysis.ipynb — Python cleaning and feature engineering notebook
- customer_shopping_behavior_analysis.sql — 15 SQL queries (10 original/corrected + 5 new, covering CTEs, window functions, subqueries, and HAVING)
- Power BI dashboard (.pbix) — interactive visualization